

MOD 3 – project report

Using clustering algorithms to assess the validity of biotope codes assigned to plant surveys

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Introduction

In times of habitat and plant biodiversity loss (Jandt et al., 2022), the restoration and management of valuable biotopes are highly relevant. As a first step, these biotopes need to be mapped, for example in military training areas, which provide large, continuous and diverse habitat networks with a heterogeneous disturbance regime (Warren et al., 2007). Such mapping includes the identification of vascular plant species and their abundance per biotope, followed by the assignment of a biotope code. Ideally, this would be carried out by specialists, however, financial constraints limit staff availability and time, leading to a decline of mapping quality.

Quality assurance could involve either the verification of each mapped plot by a specialist or the use of highly specific rule-based controls to check whether the plant composition matches the assigned biotope code. An alternative would be to automatically cluster plant data and assess whether the assigned biotope codes align with the resulting clusters. However, this approach relies on the assumption that each biotope code, or at least each main biotope category, is characterized by a unique plant composition. Therefore, the aim of this study is to investigate whether clustering algorithms can group plant data according to their assigned biotope codes by comparing three different algorithms and exploring the potential of ordination methods to support clustering and visualize the results.

Methods

The first step was preparing and cleaning the data. Although the dataset had previously been submitted to the employer, it still contained some unrealistic plots, which were removed. To further improve the performance of the clustering algorithms, the dimensionality of the data was reduced stepwise. This included splitting the data into forest (3786 plots) and grassland (298 plots), removing biotopes at early successional stages, excluding entries with only a few plant species (one or two) and finally removing mixed forest biotopes.

Plant abundance was recorded in five abundance classes, ranging from 1 (rare species) to 4 (highly abundant species), with class 9 being largely equivalent to class 4. These classes were transformed into percentage cover values using two different schemes. In the first, more balanced approach, classes 1 to 4 were transformed in 0.01, 0.05, 0.25, and 0.9, respectively. In the second approach, greater importance was assigned to highly abundant species, resulting in the transformation 0.01, 0.01, 0.01, and 1, respectively. It was assumed that the total percentage cover could exceed 100% per plot. However, to ensure comparability between plots based on species composition and to reduce the influence of overall vegetation density, the data for each plot were normalized so that the total cover summed to 1.

To determine which reduced dataset yielded the best clustering performance, datasets with increasing levels of simplification were evaluated. These included removing plots with few plant species, aggregating plant data to the genus level, considering only tree species, and excluding mixed forest biotopes. Initial tests were conducted using

HDBSCAN. HDBSCAN is a density-based clustering algorithm which builds a clustering hierarchy and requires the minimum cluster size as primary parameter (Campello et al., 2013). The algorithm operates on a distance matrix as input.

Two additional algorithms were applied: partitioning around medoids (PAM) and gaussian mixture modelling (GMM), thereby incorporating algorithms based on different underlying principles (Yin et al., 2024). PAM is related to the k-means algorithm but uses medoids, which represent actual data points, as cluster centres (Nkweteyim, 2018). PAM can also operate on a distance matrix. The optimal number of medoids must be specified, in this study, values ranging from 2 to 25 were tested. Model selection was based on the average silhouette width of which the highest value is chosen (Nkweteyim, 2018).

In contrast to the previously described algorithms, GMM operates directly on the abundance matrix and determines the optimal model automatically. This is achieved by minimizing the log-likelihood function and evaluating the Bayes Information Criterion (BIC) (Melnykov and Maitra, 2010). Prior to clustering, only transformation and normalization were applied.

To visualize the results, methods such as PCA, NMDS or UMAP were considered for dimensionality reduction. In addition, clustering algorithms can be applied on ordination data too. Therefore, the effect of dimensionality reduction using NMDS or UMAP on the clustering performance of HDBSCAN was evaluated for the best-performing dataset. The number of NMDS dimensions was determined based on visual inspection of stress plots.

Results

The original dataset consisted of 4597 plots. After removing unrealistic plots and those representing early successional biotope codes were removed, 3595 plots remained. The distribution of plots across biotope codes and biotope groups is shown in Figure 1A in the Appendix.

Both Euclidean and Bray–Curtis distance measures were tested, as the latter performed better, it was used for all subsequent analyses. Dataset evaluation was based on the adjusted Rand index (ARI), purity and the fraction of noise in case of HDBSCAN. Given the aim of this study, purity was considered the most relevant metric, as it reflects the extent to which clusters predominantly contain a single biotope group. However, ARI was also taken into account as a standard measure of clustering performance. For HDBSCAN, minimum cluster sizes ranging from 3 to 20 were tested. Based on this evaluation, five datasets, each with a corresponding transformation scheme, were selected for comparison across the three algorithms.

An advantage of HDBSCAN is its ability to classify observations as noise, allowing performance metrics to be calculated while excluding these points. However, in such cases, the fraction of noise must be considered separately. When all three algorithms were evaluated using metrics calculated across all clusters, PAM achieved the highest purity at the biotope group level for the dataset restricted to plots containing at least

one tree species, further reduced to tree species only, and excluding mixed forests. The nine resulting clusters are shown in Figure 1. Figure 1a presents the mean silhouette width and the number of observations per cluster. Clusters 1, 2 and 4 exhibit the highest mean silhouette values, all exceeding 0.6, indicating that the distance between the points in the cluster is low and the distance to points in other clusters high. This pattern is also visible in Figure 1b, where the data points - projected in two-dimensional UMAP space - are enclosed by hulls representing the PAM clusters. While some parts of these clusters are spatially separated from the rest, only few points lie in the middle. However, it should be noted that distances in UMAP space are not directly comparable to those used by PAM. Clusters that are more centrally located, such as clusters 6, 7 and 8, exhibit lower mean silhouette values. Notably, the biotope group “AB” is dominant in two clusters.

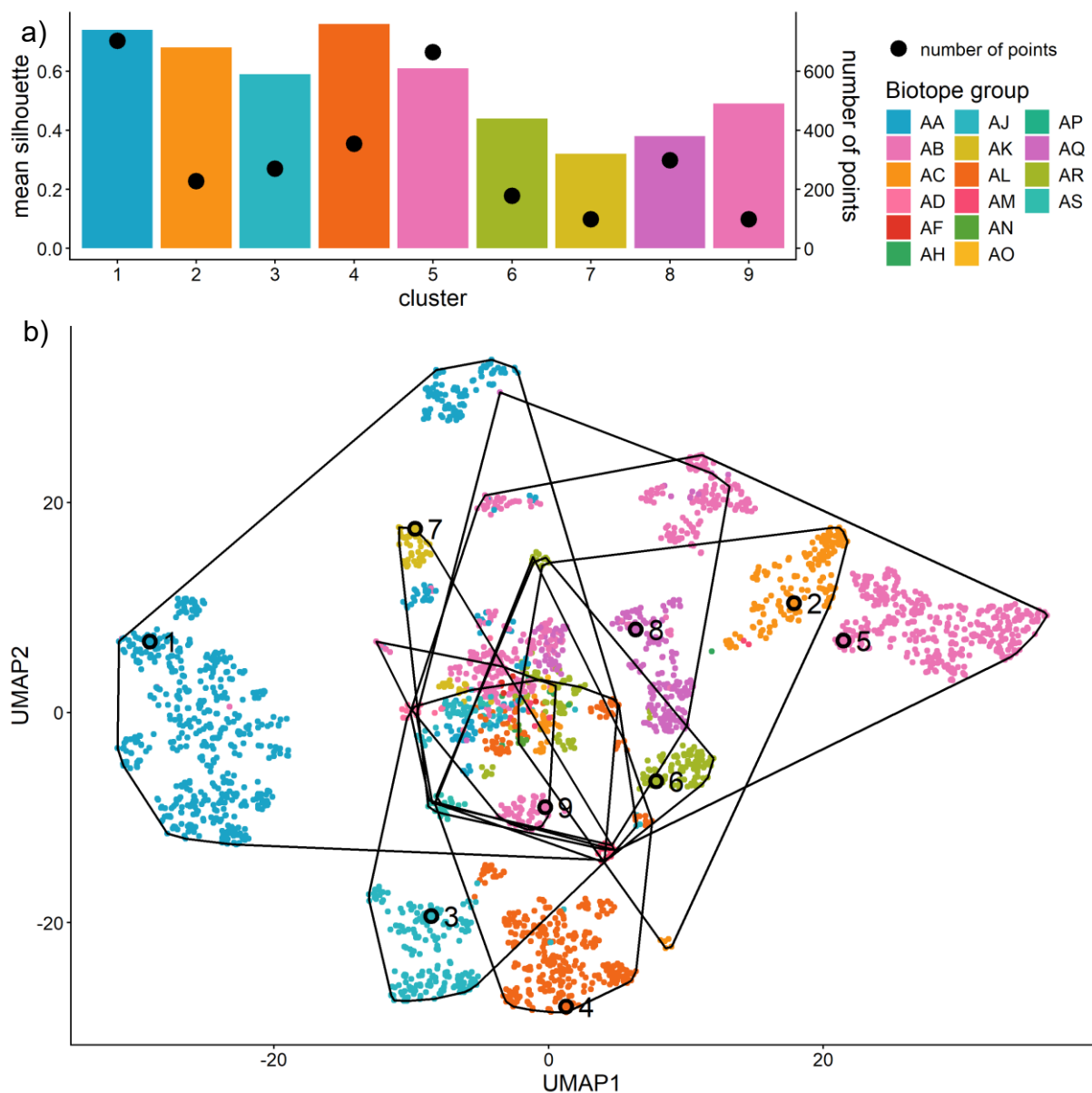


Figure 1: a) Mean silhouette width per cluster shown as bar plot, the colours represent the dominant biotope group within each cluster and the number of observations per cluster is indicated by points. b) Two-dimensional UMAP representation of the data coloured by biotope group, clusters are displayed as hulls and medoids are highlighted and labelled according to their respective clusters.

In general, GMM performed worst among the tested algorithms; for two data sets, it could not calculate clusters. Figure 2 visually compares the three algorithms as well as the performance of HDBSCAN applied to a five-dimensional NMDS representation.

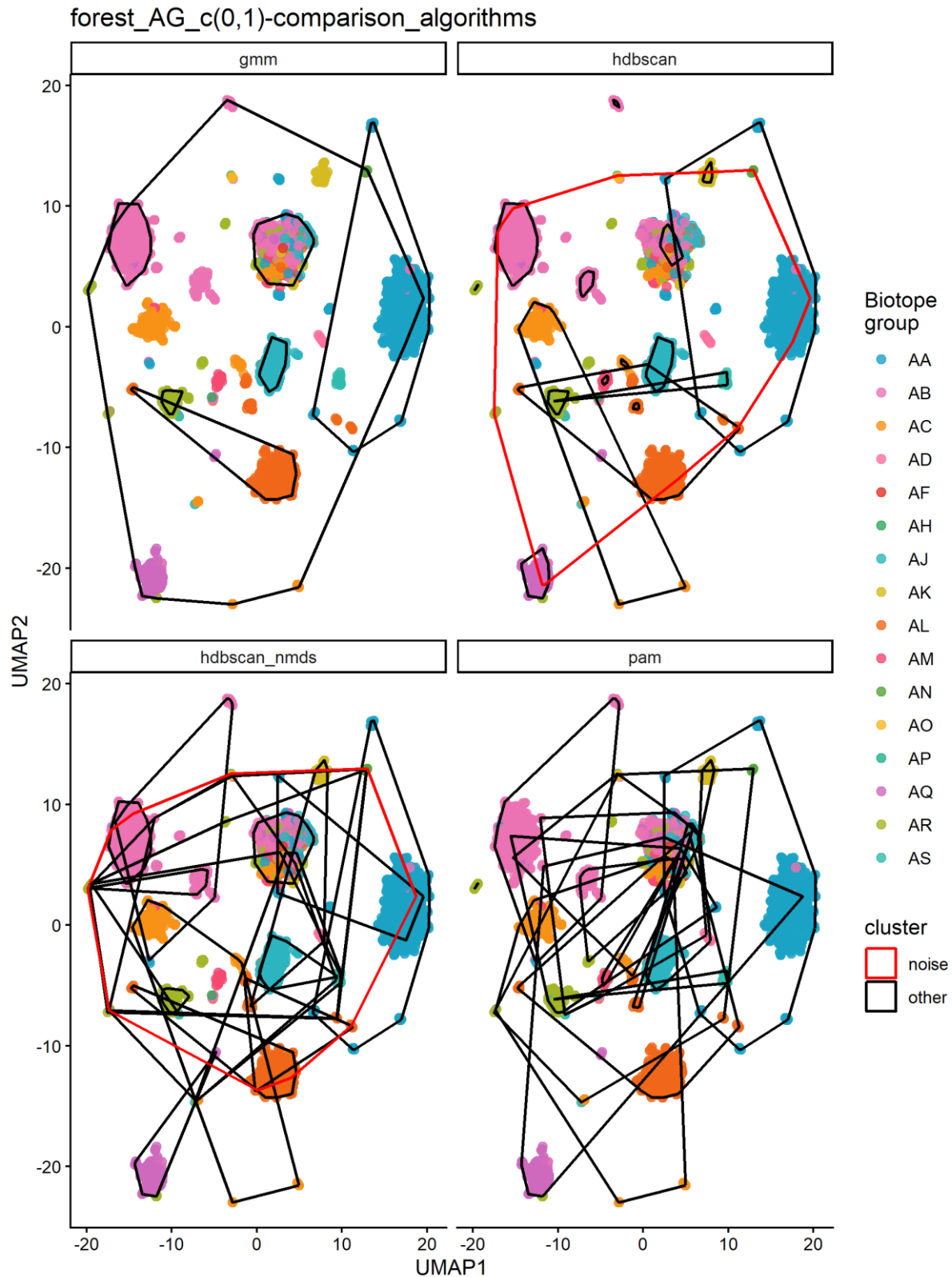


Figure 2: Two-dimensional UMAP representation of the plant datasets. Data points are coloured according to the biotope group, and hulls indicate the clusters generated by the respective algorithms.

The selected dataset included all plant data while excluding plots containing only a single plant species and all “AG” biotope codes. When purity of HDBSCAN was calculated excluding observations classified as noise, the value reached almost 0.98. The individual clusters captured the individual groups visible in the UMAP representation well and the noise points were distributed across almost the entire plot. The proportion of observations classified as noise was 23.7%. Using a minimum cluster size of 13 resulted in 17 clusters, which is only slightly higher than the number of biotope group codes. In addition, it was again apparent that some biotope groups, such as “AB”, were represented by multiple clusters. In comparison, applying HDBSCAN on five-dimensional NMDS representation substantially reduced the proportion of noise to 3.8%. Using a minimum cluster size of 8 resulted in 19 clusters, which is still comparable to the clustering obtained without ordination. This led to a lower purity (0.85), however, when the noise observations were also included in the evaluation, the purity remains comparable, but the ARI increased (0.61 to 0.55). For PAM, 16 medoids were selected, resulting in a higher overall purity of 0.85. The PAM clusters shown in Figure 2 exhibit substantial overlap and are less clearly separated than those produced by HDBSCAN, potentially because observations that would otherwise be classified as noise are instead distributed among the different clusters. GMM produced only seven clusters and performed worst across all metrics (Table 1A).

Finally, to address the original objective of the study, the clustering result with the highest purity at the biotope code level will be considered to check incoming plots. Since HDBSCAN classifies some observations as noise, with these observations excluded from the purity calculation, the highest purity was achieved using the forest dataset restricted to tree species only and a minimum cluster size of 6. However, because the fraction of noise should not be too large, the same dataset excluding plots containing only one tree species was ultimately selected. Using a minimum cluster size of 5 resulted in 71 clusters, which is close to the 65 existing biotope codes. Nevertheless, the clusters did not fully correspond to the assigned labels, partly because 20 of the biotope codes were represented by fewer than five plots. Consequently, the resulting purity was only 0.66. The distribution of biotope groups across clusters is shown in Figure 3. The noise category contained the largest number of observations and included plots from nearly all biotope codes. Furthermore, some biotope groups were distributed across many clusters, such as “AB” which dominated many small clusters. In contrast, “AC” and “AL” - dominated by alder and Douglas fir, respectively, two tree species that occur only rarely in other biotope groups—were each concentrated mainly within a single cluster. However, even the smaller clusters, for example those dominated by “AB”, were not completely pure with respect to biotope code (Figure 3b).

Discussion

Compared with NMDS, UMAP produced the visually most intuitive two-dimensional representation, as its parameters can be adjusted more easily and have a strong influence on the resulting projection. Regarding the visual output, the minimum

distance parameter primarily determines how densely packed the displayed clusters appear (McInnes et al., 2020).

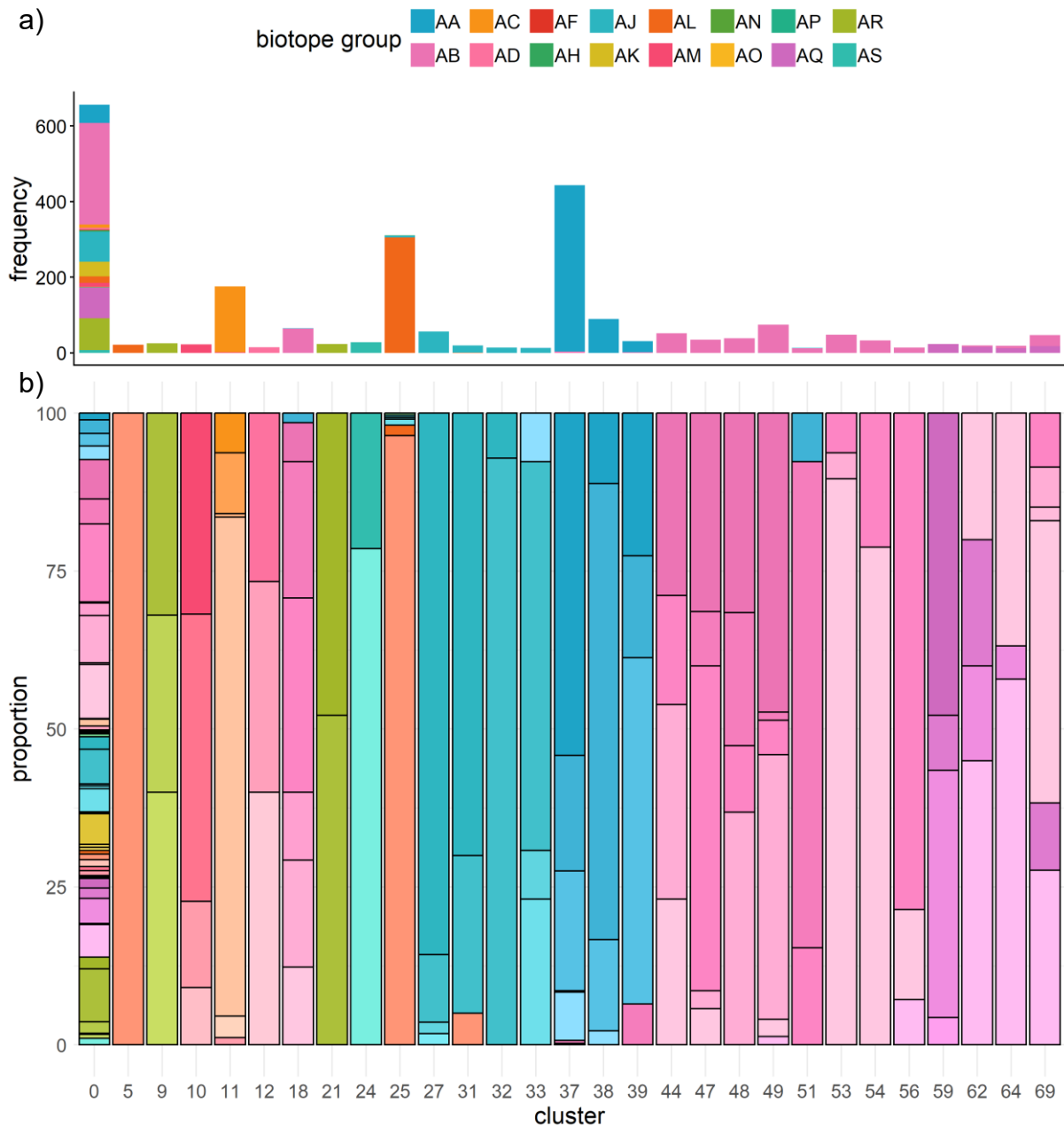


Figure 3: Clusters containing more than 12 observations resulting from HDBSCAN applied to the tree dataset excluding plots with only one tree species and “AG” biotopes. (a) Number of observations per cluster coloured by biotope group. (b) Proportion of individual biotope codes within each cluster, biotope codes belonging to the same biotope group are displayed in different shades of the corresponding group colour indicated in the legend.

All three algorithms behaved differently and produced different results. GMM performed worst overall, potentially because it does not rely on the precomputed ecological distance matrix. In contrast, the other two algorithms depend on the chosen distance measure and may have performed better by trying different distance metrics, as the choice of distance metric is generally an important factor (Nkweteyim, 2018). However, Bray-Curtis is often considered suitable for heterogeneous vegetation data (Pakgohar et al., 2021) and outperformed Euclidean distance in the initial comparison conducted in this study.

PAM performed comparatively well. PAM is strictly distance-based and has the advantage that the resulting medoids could be examined directly to assess whether they represent idealized examples of the biotope type dominating the respective clusters, thereby potentially increasing the understanding of the resulting clusters.

Furthermore, the average silhouette score provides an easily interpretable metric for selecting the number of medoids; however, it does not guarantee that the optimal number of clusters will be found (Nkweteyim, 2018). In addition, the missing of an explicit classification of noise observations is disadvantageous in cases such as the present study, where the data is not perfect.

HDBSCAN clusters observations based on density by considering core distances, while observations that are not core objects are treated as noise (Campello et al., 2013). This increases the purity of the remaining clusters. When using a relatively large minimum cluster size, even the last cluster contained highly mixed biotope codes. Therefore, both this and the noise observations were interpreted as potential outliers, which is consistent with manual inspection, as some plots were likely misclassified. A second clustering run using only these observations and alternative weighting schemes did not yield satisfactory results.

Consequently, incoming plots could be predicted into one of the clusters of the HDBSCAN model shown in Figure 3. The assigned biotope code could then be compared with the proportion of that code in the respective cluster. Based on this comparison, a threshold for sufficient agreement could be defined. Furthermore, if the assigned biotope code itself is not represented within the cluster, the same comparison could still be performed at the broader biotope group level. Depending on available staff and time capacities, it could then be decided whether agreement on biotope group level is sufficient for quality assurance purposes. In addition, membership probabilities could also be incorporated into the decision-making.

Conclusion

Overall, the possibilities for conducting cluster analyses are seemingly overwhelming, ranging from data transformation and normalization to the selection of distance measures, algorithm parameters and the used (combination of) evaluation metrics. In addition, the possibilities are extended by the use of different ordination methods, both for potentially improving clustering performance and for visualizing the results. Therefore, a clearer strategy regarding which methodological aspects to explore and the early definition of a benchmark against which all subsequent results could be compared, would have been useful. Furthermore, a stronger theoretical understanding of the applied methods (including github) would likely have streamlined the work and potentially improved the results. Nevertheless, the final results appear suitable for evaluating newly mapped plots. In addition, the analysis enabled the identification of many potentially misclassified plots that can now be reviewed. Remaining questions include whether a hard-coded rule-based control system would be more time-efficient and whether classifier, such as random forest, could achieve better performance, potentially when trained only on observations excluding the HDBSCAN noise category.

Appendix

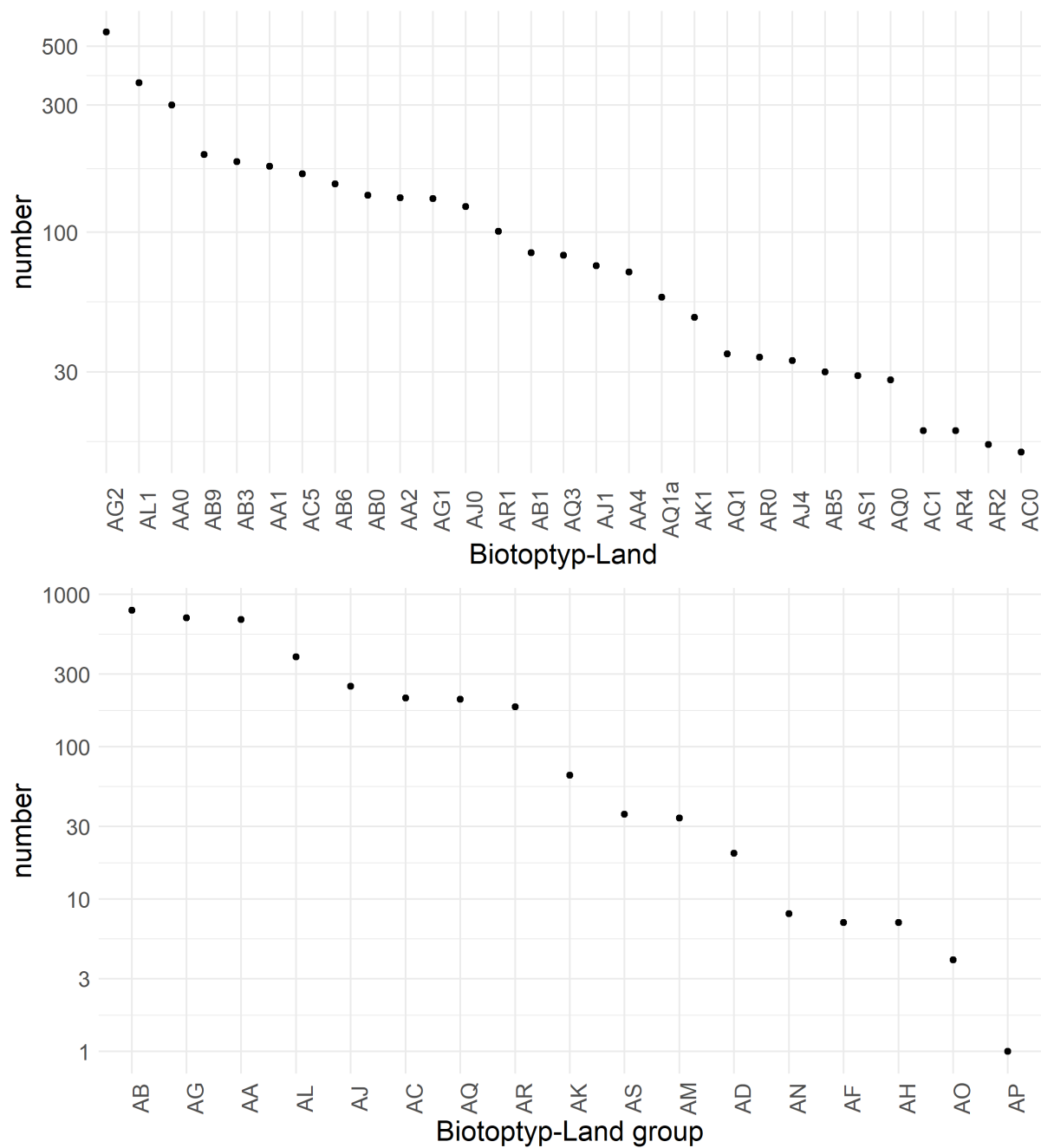


Figure 1A: Number of plots per biotope code for all codes represented by more than 14 plots (top) and number of plots per biotope group (bottom).

Table 1A: Parameters and evaluation metrics of HDBSCAN, PAM, and GMM for the five selected datasets. “_” indicates reductions or simplifications of the dataset, numbers in brackets indicate that plots containing only up to the specified number of plant species were removed. “-all” indicates that observations classified as noise were included in the metric calculation. “combined” represents the combined score calculated as $0.5 \times ARI + 0.5 \times \text{purity}$.

		Forest-AG- (0,1)	Trees- genus-AG- (0,1)	Trees-AG	Trees-AG- (0,1)	Forest- (0:2)
H D B S C A N	Minimum cluster size	13	7	6	5	20
	Number of clusters	17	23	83	71	13
	Noise	683	560	843	656	1218
	Biotope code purity	0.53	0.52	0.69	0.66	0.52
	Biotope code_purity-all	0.43	0.44	0.53	0.53	0.46
	Biotope group purity	0.98	0.96	0.96	0.96	0.94
	Biotope group purity-all	0.82	0.84	0.77	0.83	0.76
	Biotope group ari-all	0.55	0.62	0.12	0.34	0.42
	Biotope group-combined	0.68	0.73	0.45	0.59	0.59
P A M	Medoids	16	14	9	9	24
	Biotope code purity	0.45	0.45	0.45	0.44	0.49
	Biotope group purity	0.85	0.87	0.88	0.88	0.81
	Biotope group-combined	0.77	0.8	0.82	0.82	0.67
G M M	Number of clusters	7	8	1	1	8
	Biotope code purity	0.36	0.38	0.13	0.12	0.43
	Biotope group purity	0.68	0.73	0.27	0.28	0.7
	Biotope group-combined	0.57	0.65	0.14	0.14	0.58

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